CHAPTER 4 · LESSON 3: MACHINE & NATURE ALGORITHMS (Pages 46–48)

■ **One-Line Topic Concept:** Washing machines follow built-in computer algorithms, and growing trees follows nature's biological step-by-step algorithm!

Activity 1 · Number the Steps for Planting a Sapling (Textbook Page 46):

[**4**] Water the plant regularly and provide sunlight. ➔ *(Step 4)*

[**1**] Dig a small pit in fertile garden soil. [START] ➔ *(Step 1)*

[**5**] Watch the sapling grow into a healthy tree! [STOP] ➔ *(Step 5)*

[**2**] Place the young sapling carefully into the pit. ➔ *(Step 2)*

[**3**] Cover the roots properly with soil and compost. ➔ *(Step 3)*

Activity 2 · Washing Machine Algorithm: Fill in the Steps (Textbook Page 47):

Word Bank: Detergent • START • Spin & Dry • Dirty Clothes • STOP

• Step 1: Press the power button to **START** the machine.

• Step 2: Put the **Dirty Clothes** inside the washing drum.

• Step 3: Add washing powder / **Detergent** into the dispenser tray.

• Step 4: Machine washes, rinses, and enters **Spin & Dry** cycle.

• Step 5: Machine beeps, take out clean clothes and **STOP**.

Activity 3 · Number the Steps for a Tailor Stitching a Shirt (1 to 4):

(Based on Textbook Page 47 Tailor Algorithm):

[**3**] Stitch the cut pieces together using the sewing machine. ➔ *(Step 3)*

[**1**] Take body measurements using a measuring tape. [START] ➔ *(Step 1)*

[**4**] Attach buttons, iron the shirt, and pack neatly! [STOP] ➔ *(Step 4)*

[**2**] Mark and cut the cloth fabric according to measurements. ➔ *(Step 2)*

Activity 4 · Answer in Complete Sentences (Spacious Writing Lines):

1. Why does a washing machine need an exact algorithm to work properly?

➔ ***Because it is an automatic machine that must wash, rinse, and spin dry in exact sequence.***

2. What biological algorithm does a tiny seed follow to grow into a tall green tree?

➔ ***Roots absorb water and nutrients from soil while stem grows upward toward sunlight to sprout leaves.***

Activity 5 · Real-Life Algorithm Comparison Master Table:

Category	Starting Step (Input)	Step-by-Step Algorithm (Process)	Final Goal (Output)
Machine Algorithm (Washing Machine)	Press Power & load dirty clothes	Washes with soap, rinses with water, and spins dry automatically	Clean, dry clothes ready to fold
Nature Algorithm (Planting a Sapling)	Dig fertile pit and place seedling	Roots drink water from soil; leaves absorb sunlight to grow stem taller	Strong green tree with shade & fruits
Human Algorithm (Tailor Stitching)	Measure student with measuring tape	Cut cloth fabric neatly along markings and stitch seams on machine	Neat, perfectly fitted school uniform